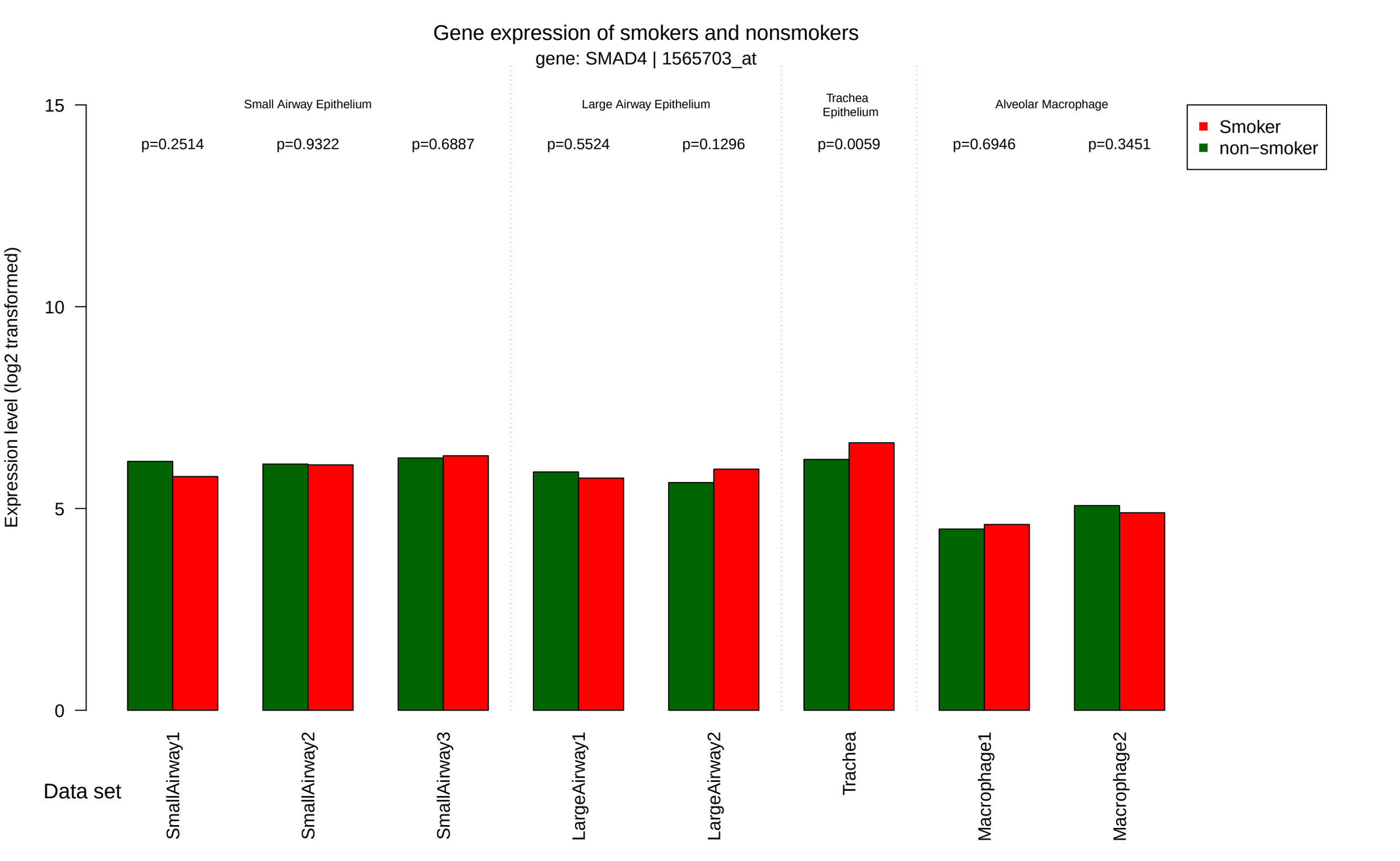
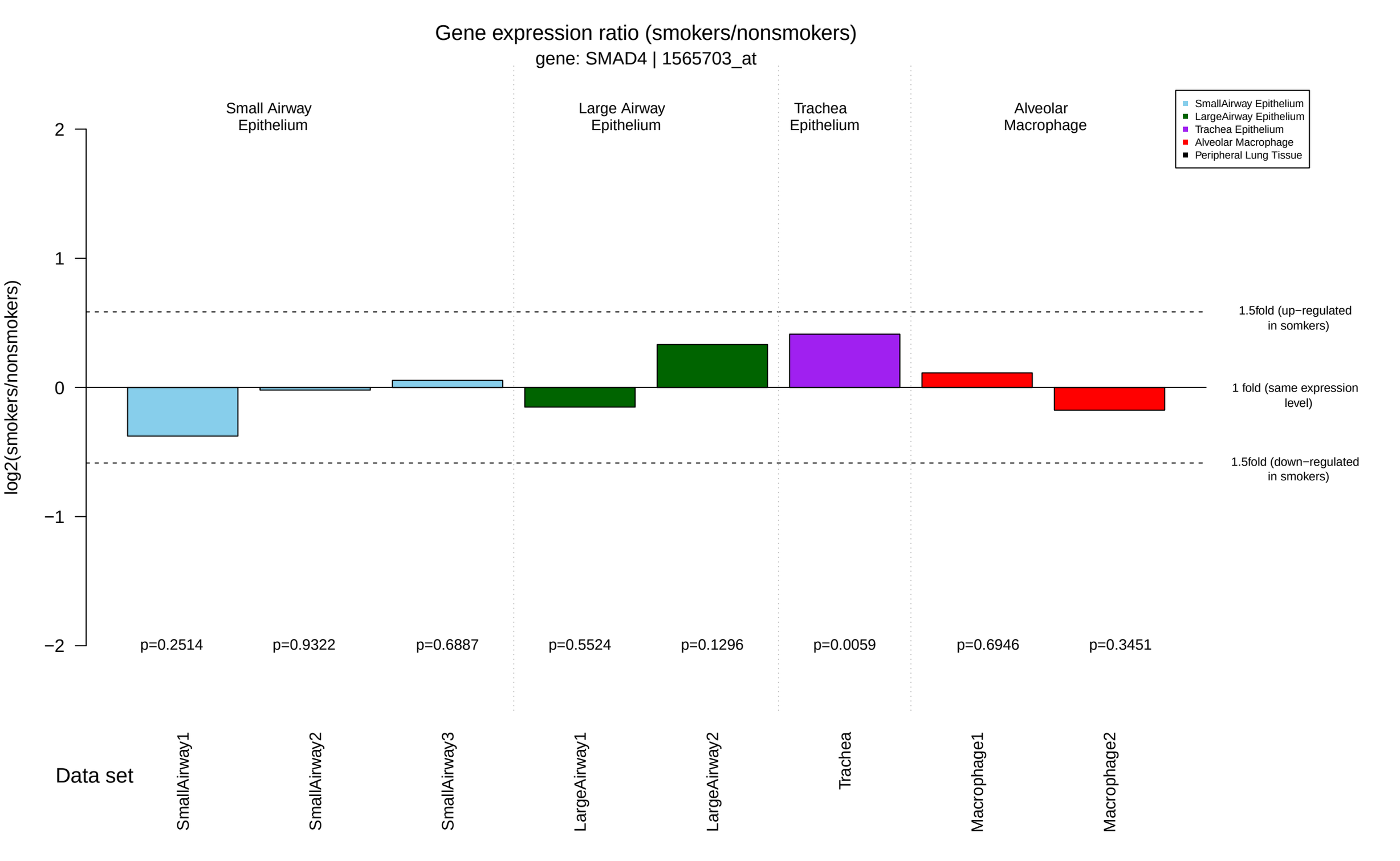
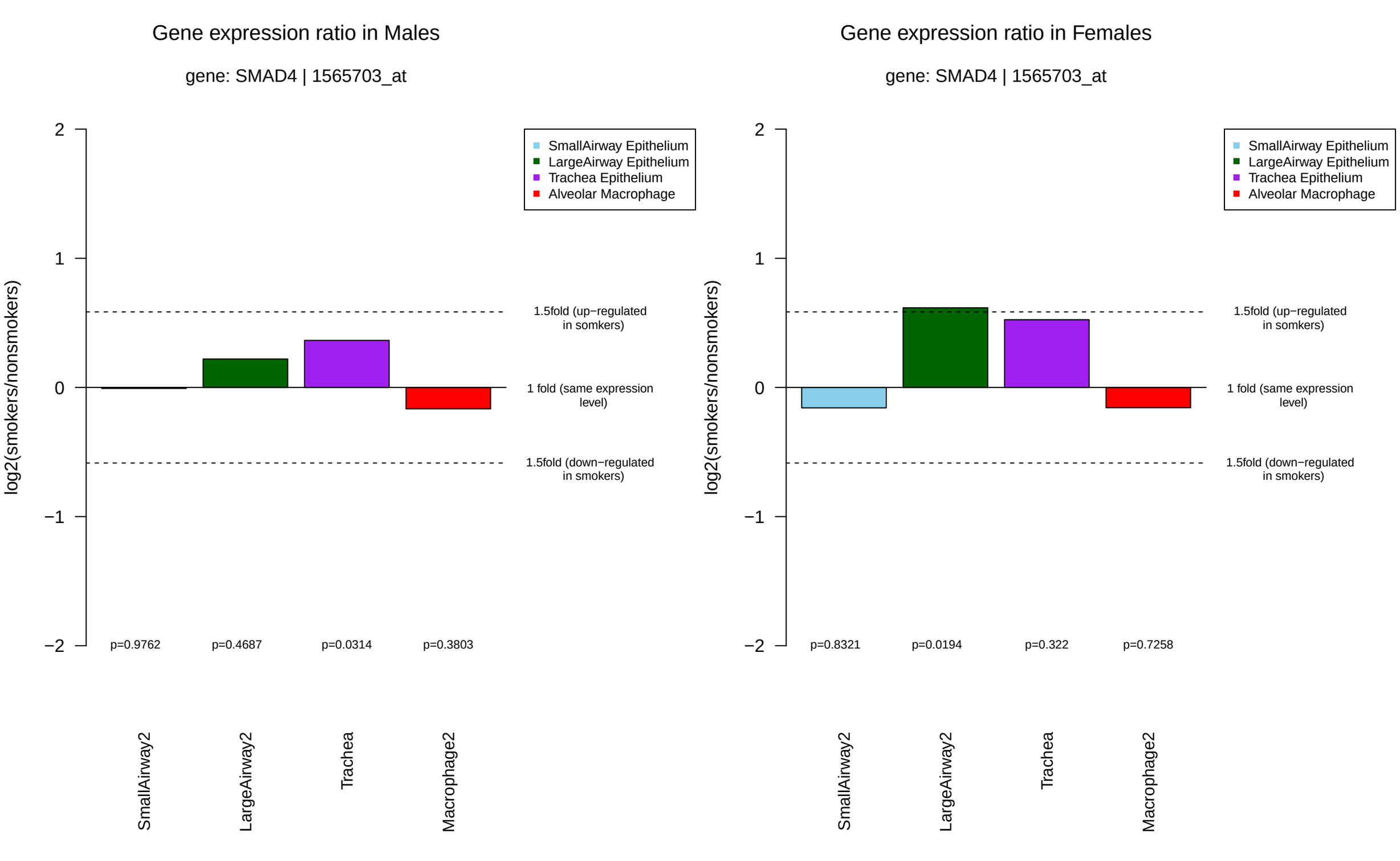
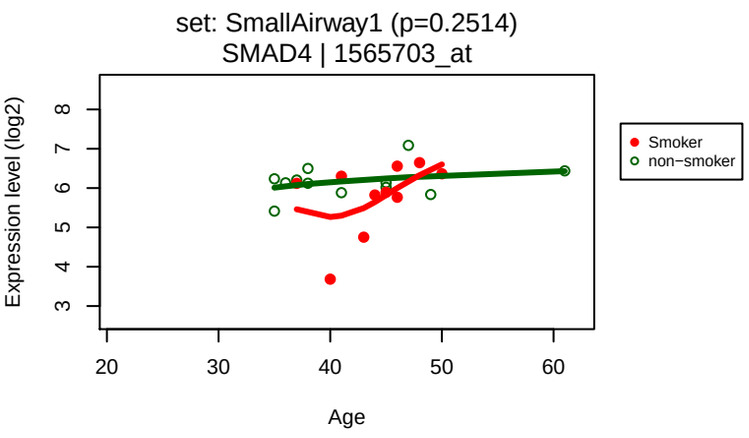
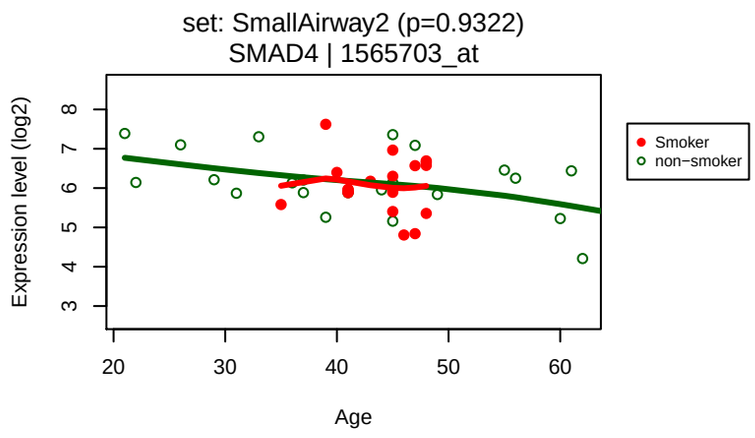
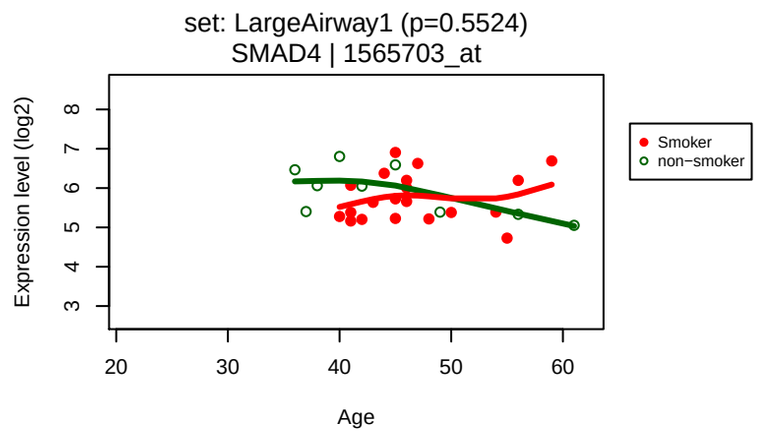

Gene expression of smokers and nonsmokers
gene: SMAD4 | 1565703_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2514
p=0.9322
p=0.6887
p=0.5524
p=0.1296
p=0.0059
p=0.6946
p=0.3451
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

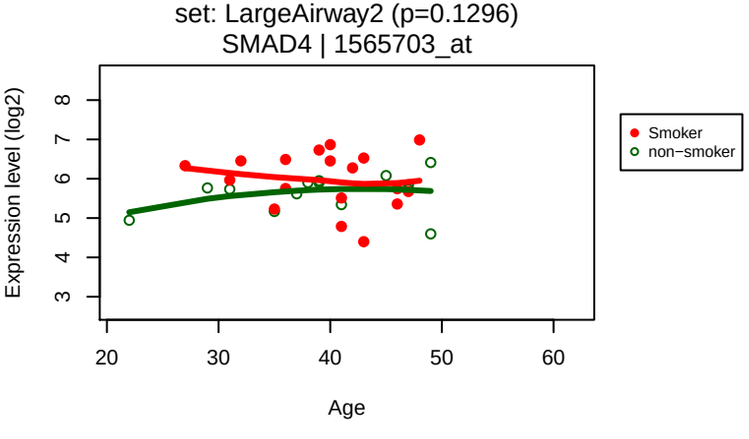

Gene expression ratio (smokers/nonsmokers)
gene: SMAD4 | 1565703_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2514
p=0.9322
p=0.6887
p=0.5524
p=0.1296
p=0.0059
p=0.6946
p=0.3451
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

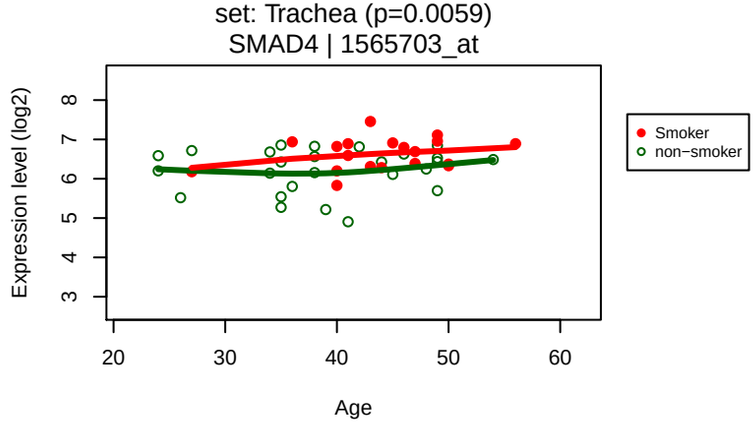

Gene expression ratio in Males
gene: SMAD4 | 1565703_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9762
p=0.4687
p=0.0314
p=0.3803
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMAD4 | 1565703_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8321
p=0.0194
p=0.322
p=0.7258
SmallAirway2
LargeAirway2
Trachea
Macrophage2

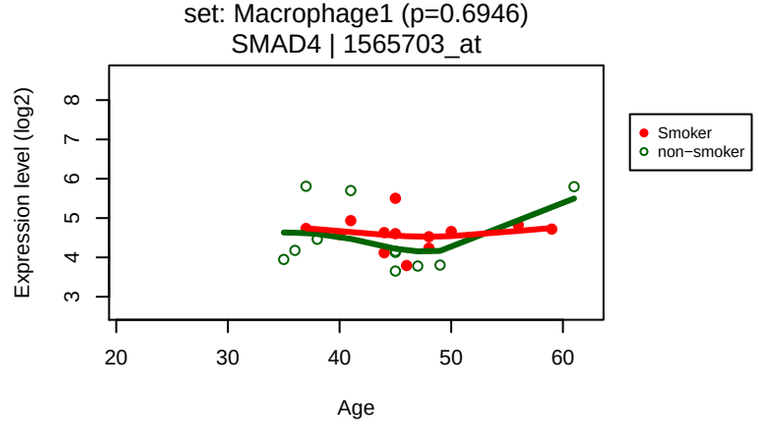

set: SmallAirway1 (p=0.2514)
SMAD4 | 1565703_at
Expression level (log2)
Age
Smoker
non−smoker

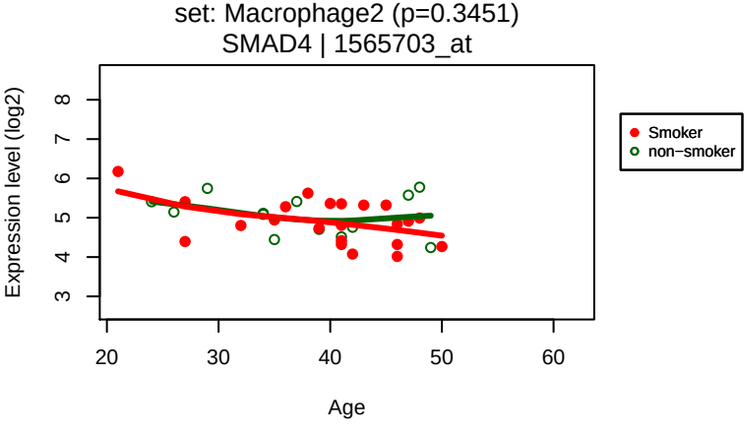

set: SmallAirway2 (p=0.9322)
SMAD4 | 1565703_at
Expression level (log2)
Age
Smoker
non−smoker

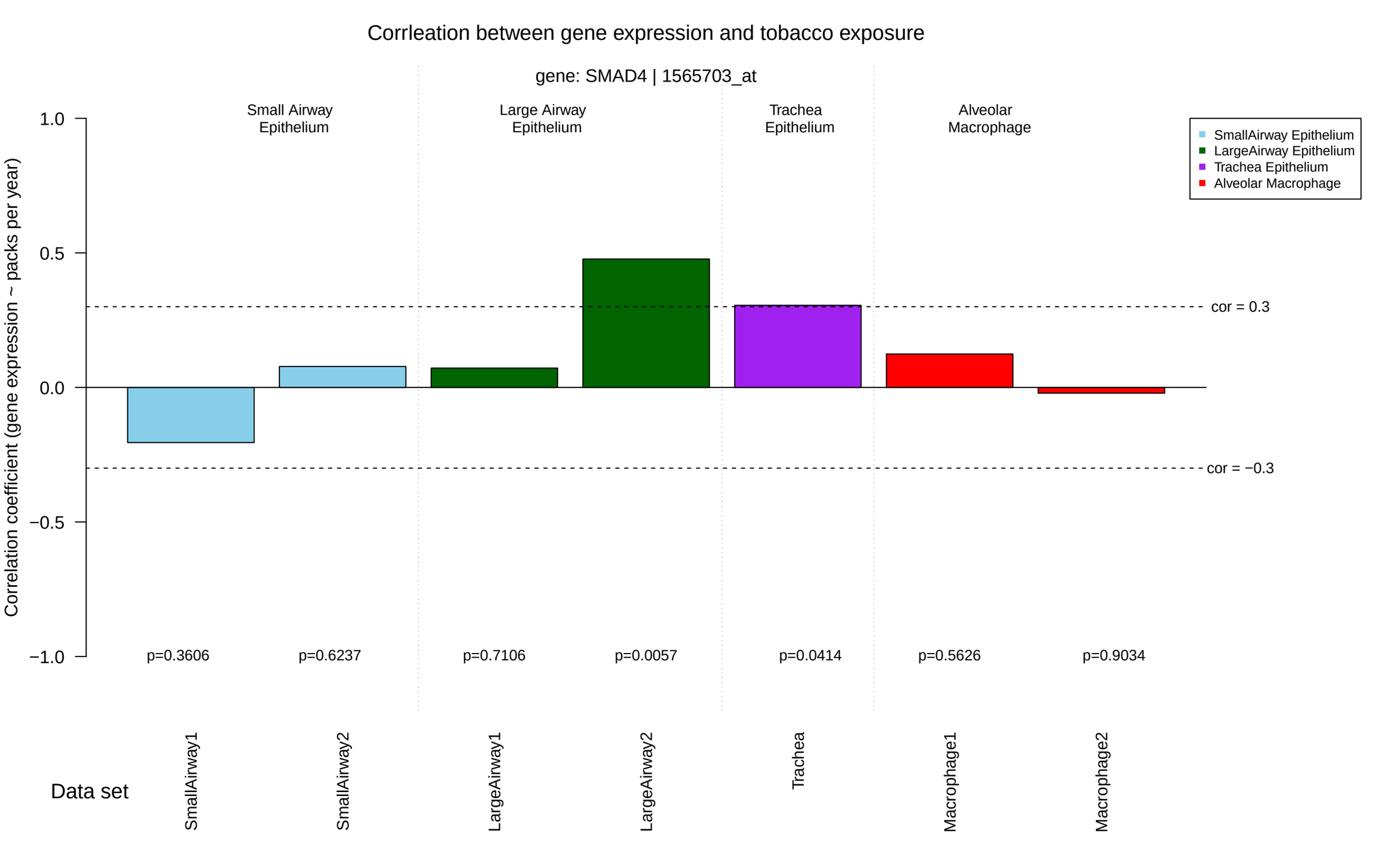

set: LargeAirway1 (p=0.5524)
SMAD4 | 1565703_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1296)
SMAD4 | 1565703_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0059)
SMAD4 | 1565703_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6946)
SMAD4 | 1565703_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3451)
SMAD4 | 1565703_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMAD4 | 1565703_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3606
p=0.6237
p=0.7106
p=0.0057
p=0.0414
p=0.5626
p=0.9034
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2